ESMC-6B State 51 Weight Geometry

A weights-only investigation of the ESMFold2 mixing outlier

Bottom line

State 51 receives approximately 10% of the learned ESMFold2 state-mixing mass in all four supported checkpoints. The relevant producer is transformer block 50. The strongest weights-only explanation is a projection-aligned write/relay event: block 50 reorganizes multiple attention heads and its SwiGLU feed-forward pathway into compact directions preferentially read by the shared 256-dimensional folding projection. Generic contact, SAE, intrinsic-dimension, norm, and compression criteria need not reward those same directions.

Analysis scope	All 80 ESMC transformer blocks; four ESMFold2 checkpoints
Evidence type	Checkpoint weights only; no sequences or activations
Compute	NVIDIA GH200 480GB workstation
ESMC revision	`45b0fa5d7fb06faefbd5e3b89bdcef35d564e79a`
Validation	64/64 saved data-quality checks passed
Report date	July 27, 2026

Abstract

This report analyzes the checkpoint geometry of all 80 transformer blocks in ESMC-6B and the mixing/projection weights of four ESMFold2 checkpoints. State 51 is a reproducible mixer outlier, receiving 9.988%–10.654% of total softmax mass and ranking fourth in every checkpoint. Correct indexing maps this state to the output of block 50. Block 50 exhibits statistically unusual alignment between compact feed-forward input subspaces and the folding projection, a network-wide maximum in the number of strongly coupled attention heads, and a sharp redistribution of SwiGLU neuron coupling. These effects persist under scale and LayerNorm sensitivity checks. In contrast, weight-vector intrinsic dimension, normalization statistics, and weight-only compression do not robustly isolate blocks 50–51. The evidence supports a task-specific, projection-readable interface state, but does not establish activation usage, folding causality, or biological meaning.

Contents

1 Question, evidence, and conclusion

The motivating discrepancy is straightforward. Contact-oriented P@L performance and the SAE curves in Figure S26 of the supplied preprint dip near layer 51, yet the ESMFold2 mixer assigns that state unusually high mass. The question is whether the checkpoint weights contain a structural explanation for why a folding model would prefer a state that appears weaker under other representational criteria.

Three observations must be kept separate:

- P@L measures whether residue-contact information is accessible to a particular probe or attention-derived statistic.
- The Figure S26 SAE metrics measure reconstruction difficulty and masked-language-model perplexity change after SAE reconstruction.
- ESMFold2 mixing measures which normalized and projected states are useful to a jointly trained folding objective.

These are different functionals of a hidden state. A low-volume direction can matter strongly to a 256-wide folding projection while contributing little to average reconstruction loss, masked-token prediction, or contact precision. The weight analysis tests whether such a pathway is geometrically available. It cannot establish that real protein activations occupy that pathway.

Primary conclusion

The positive evidence localizes to block 50, the producer of state 51. It is a coordinated event spanning FFN projection alignment, multi-head V–O coupling, and neuron-wise FFN redistribution. Block 51, the consumer of state 51, is not a comparably robust isolated anomaly.

2 Model interface and the indexing distinction

ESMC exposes 81 ordered hidden states from 80 transformer blocks:

$$\begin{aligned} h_0 &: \text{ token embedding state,} \\ h_1 &: \text{ output of block 0,} \\ h_s &: \text{ output of block } s-1, \\ h_{51} &: \text{ output of block 50,} \\ h_{80} &: \text{ output of block 79.} \end{aligned}$$

Consequently, state 51 is produced by block 50 and consumed by block 51. Calling the feature "layer 51" without distinguishing state and block indices shifts the putative causal location by one block. Both blocks were prespecified, together with blocks 48–53 and late-layer controls 76–79.

For learned mixing logits c_s, LayerNorm N, shared bias-free projection $W \in \mathbb{R}^{256\times 2560}$, and state vector $h_s \in \mathbb{R}^{2560}$, the implemented folding input is

$$\alpha_s = \frac{\exp(c_s)}{\sum_{t=0}^{80}\exp(c_t)}, \tag{1}$$

$$z = \sum_{s=0}^{80} \alpha_s W N(h_s). \tag{2}$$

There is one shared projection per checkpoint, not a separate matrix for each state. Projection precedes mixing. Because LayerNorm is nonlinear, Equation 2 is not generally equivalent to normalizing a raw weighted sum of hidden states.

The coefficient $\alpha_{51} \approx 0.10$ means that state 51 receives about one tenth of the convex mixing mass. It does not imply one tenth of the final information, norm, variance, or causal contribution. Projected states can correlate, cancel, or be redundant.

3 Mixer consistency across four ESMFold2 checkpoints

Table 1: Exact mixing summaries. Mass values are fractions of the 81-state softmax total.

Checkpoint	α_{51}	Rank	Mass 77–80	Top-1	Effective states
ESMFold2	0.10654	4	0.65412	0.23081	10.64
Experimental 2025	0.09988	4	0.60439	0.21658	11.46
Experimental Fast 2025	0.10253	4	0.62210	0.24570	10.86
ESMFold2 Fast	0.10134	4	0.63480	0.24293	10.79

The pairwise Pearson correlations between the full 81-state profiles range from 0.988 to 0.997, with Spearman correlations from 0.981 to 0.996. The last four states carry 60.44%–65.41% of the total mass, but a distinct state-51 peak remains in every checkpoint.

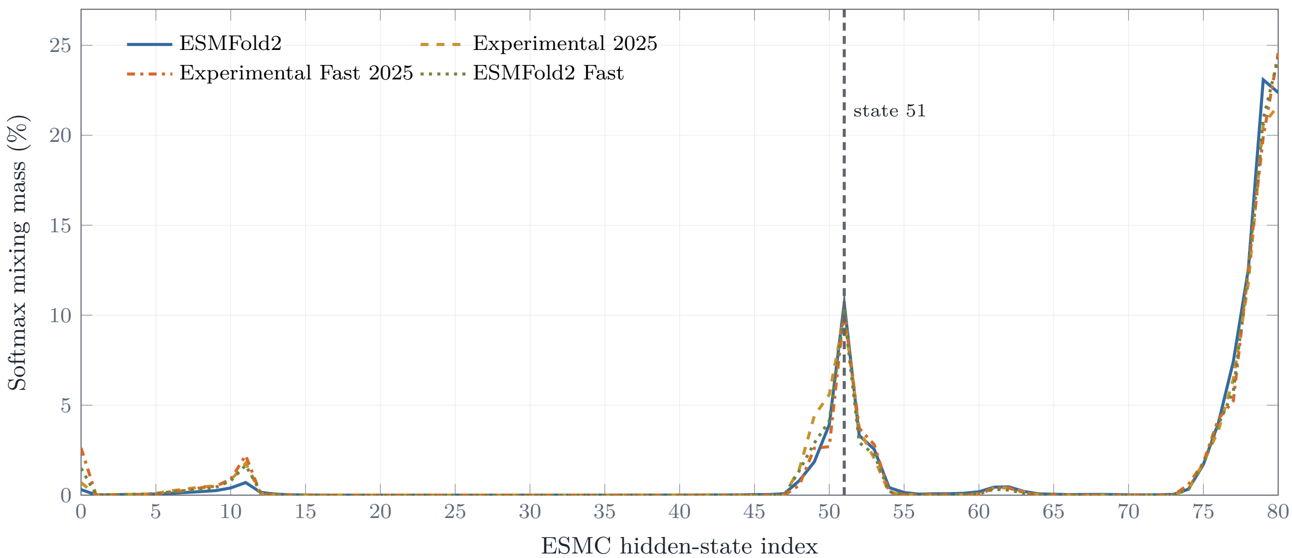

Figure 1: All four ESMFold2 checkpoints strongly favor late states while preserving an earlier state-51 peak. The checkpoint agreement is a robustness result, not four independent biological replicates.

The projection matrices themselves are not duplicates. Pairwise raw projection Frobenius cosines are only 0.121–0.228, rank-16 row-subspace overlaps are 0.345–0.451, and rank-256 overlaps are 0.263–0.339. Thus the checkpoints read related but materially different 256-dimensional subspaces while converging on almost the same depth profile.

4 Projection alignment: the strongest direct explanation

Let $U_r \in \mathbb{R}^{2560\times r}$ be a rank-r orthonormal basis from an ESMC weight matrix and let $P \in \mathbb{R}^{2560\times 256}$ span the row space of the folding projection. The normalized overlap is

$$\mathcal{O}_r(U,P) = \frac{\|U_r^\top P\|_F^2}{r} = \frac{1}{r}\sum_{i=1}^{r}\cos^2\theta_i, \tag{3}$$

where θ_i are principal angles. The score is one for full containment and zero for orthogonality. For isotropically random subspaces in 2560 dimensions,

$$\mathbb{E}[\mathcal{O}_r] = \frac{256}{2560} = 0.10. \tag{4}$$

The comparison used input-side right singular vectors for Q, K, V, gate, and value matrices, and residual-output left singular vectors for O and the FFN down projection. Ranks 16, 32, 64, 128, and 256 were tested with both raw and LayerNorm-scaled folding projections.

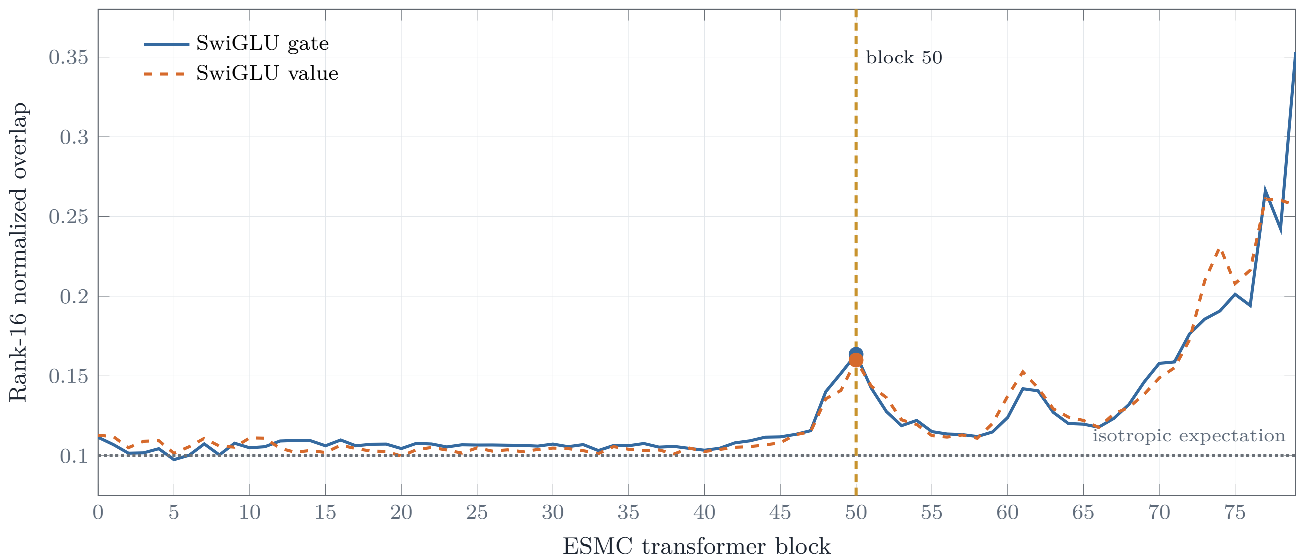

Figure 2: Four-checkpoint mean rank-16 overlap between ESMC FFN input subspaces and the folding projection. Block 50 is a local peak for both gate and value directions.

Table 2: Prespecified block-50 projection-alignment results.

Family	Sensitivity	Overlap	Local robust z	Null p	BH q
Value, rank 16	Raw	0.15995	2.46	0.0003	0.0240
Value, rank 16	LN-scaled	0.15872	2.32	0.0005	0.0400
Value, rank 32	Raw	0.15266	1.74	0.0006	0.0480
Gate, rank 16	Raw	0.16359	4.07	0.0033	0.1320

Every checkpoint has a positive block-50 local delta. For value rank 16 the raw deltas are +0.0288 to +0.0431, and the LayerNorm-scaled deltas are +0.0280 to +0.0423. Gate rank 16 has deltas from +0.0340 to +0.0519. The value result survives the prespecified family-wise correction; the gate result is strong supporting evidence but not an independent discovery at 5% FDR.

Across all 80 blocks, gate-projection alignment tracks the mixer after removing smooth depth trends. Mean depth-detrended Pearson correlations are 0.823, 0.814, and 0.797 at gate ranks 16, 32, and 64. Corresponding values include 0.787 for value rank 256, 0.765 for Q rank 256, and 0.720 for K rank 256. For gate rank 16 the checkpoint-specific range is 0.786–0.845, with an exact circular-shift $p = 0.0125$ in each checkpoint, the smallest possible value with 79 nonzero shifts.

Geometric intuition

The mixer favors states whose producer blocks expose compact directions compatible with the downstream folding readout. Block 50 is a local maximum of that compatibility. A direction can be small relative to the full 2560-dimensional residual stream yet disproportionately important after projection to 256 dimensions.

5 Attention geometry: a distributed block-50 event

Q, K, and V were split into 40 heads of shape 64×2560, and O into 40 corresponding 2560×64 blocks. Per-head metrics included Frobenius and spectral norms, stable and effective rank, Q–K principal angles, V–O subspace coupling, projection overlap, within-layer redundancy, and adjacent-layer similarity.

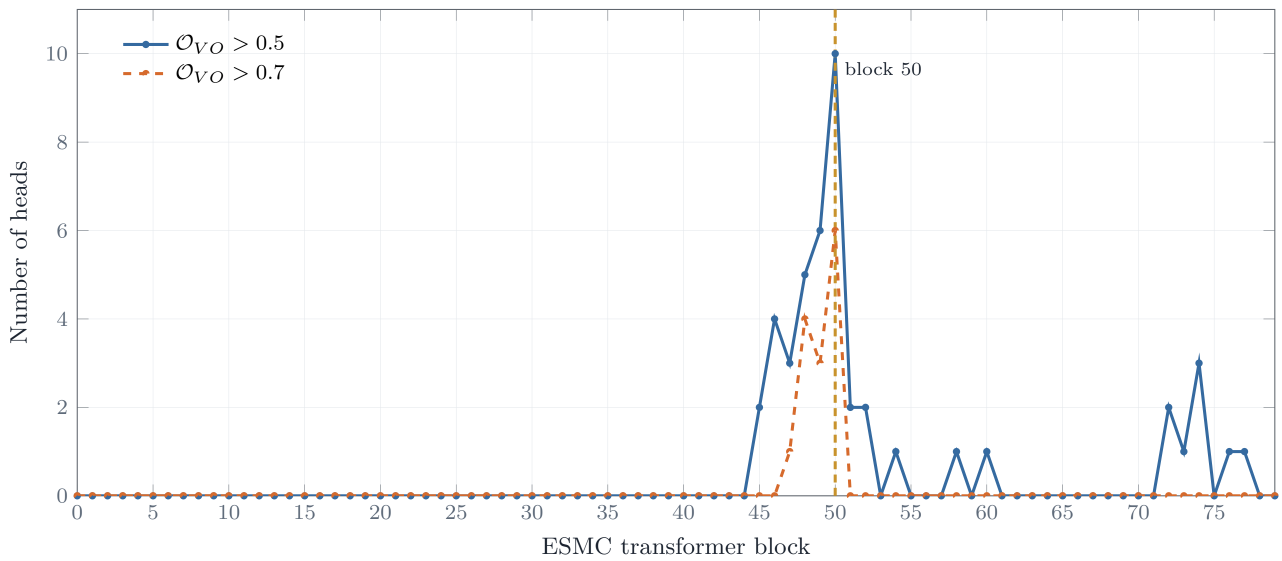

Figure 3: Counts of high V–O coupling heads among 40 heads per block. Block 50 reaches the global maximum at both thresholds and immediately relaxes at block 51.

At block 50, 10 heads exceed V–O overlap 0.5 and six exceed 0.7. These are global maxima, with BH-adjusted $q = 0.008$ and $q = 0.004$, respectively. V–O overlap dispersion is 0.25668 ($q = 0.008$), and mean O stable rank is 31.7635 ($q = 0.008$). Block 51 has only two heads above 0.5 and none above 0.7.

Table 3: Most strongly V–O coupled heads in block 50.

Head	V–O overlap	Q spectral norm	Interpretation
16	0.818	7.28	extreme coupling
20	0.814	14.34	extreme coupling
22	0.784	—	extreme coupling
38	0.779	—	extreme coupling
39	0.765	—	extreme coupling

Direct per-head projection overlap does not survive detrended significance testing. Layer-mean correlations are only moderate, approximately 0.39–0.57 with $p \approx 0.06$–0.10. The stronger result appears after aggregating head behavior and incorporating the FFN/projection pathway, arguing against a single specialized "folding head."

Hungarian matching over joint QKVO similarity identifies a local role reorganization. Transitions $49 \to 50$, $50 \to 51$, and $51 \to 52$ have matched similarities 0.09524, 0.09322, and 0.08791, ranking first, second, and eighth. Their gains over fixed head indices rank second, fourth, and third. Absolute similarities are low because full head operators change substantially with depth; the informative quantity is the relative ranking and the gain from allowing head permutation.

6 Feed-forward geometry: redistribution rather than amplification

Each block has 6912 SwiGLU neurons. Gate and value rows were paired with the corresponding FFN down-projection column. The analysis measured gate–value cosine, gate–down cosine, value–down cosine, norm ratios, subspace overlap, neuron redundancy, and triple-strength concentration.

At block 50:

- mean gate–down cosine is -0.008076, the global minimum and a local robust $z = -4.91$ with BH $q = 0.008$;
- value–down cosine standard deviation is 0.31397 with BH $q = 0.004$;
- gate–down cosine standard deviation is 0.30242 with BH $q = 0.010$.

The mean is numerically tiny, so "anti-alignment" would be an overstatement. The robust effect is a broadened neuron-wise coupling distribution, with more neurons in both unusually aligned and unusually anti-aligned tails.

This is consistent with heterogeneous routing directions, a subset of which becomes accessible to the folding projection.

7 Spectra and linear dimension

For a matrix W with singular values $s_1 \geq \cdots \geq s_m$, define

$$p_i = \frac{s_i^2}{\sum_j s_j^2}, \tag{5}$$

$$\mathrm{srank}(W) = \frac{\|W\|_F^2}{\|W\|_2^2}, \tag{6}$$

$$\mathrm{PR}(W) = \frac{(\sum_i s_i^2)^2}{\sum_i s_i^4}, \tag{7}$$

$$\mathrm{erank}(W) = \exp\left(-\sum_i p_i \log p_i\right), \tag{8}$$

$$E_1(W) = \frac{s_1^2}{\|W\|_F^2}, \tag{9}$$

$$r_\tau = \min\left\{r : \sum_{i \leq r} p_i \geq \tau\right\}, \tag{10}$$

$$\epsilon_r = \sqrt{1 - \sum_{i \leq r} p_i}, \tag{11}$$

$$C_r = \frac{r(m+n)}{mn}. \tag{12}$$

Here ϵ_r is truncated-SVD relative Frobenius error and C_r is the rank-r factorized storage fraction for an $m \times n$ matrix.

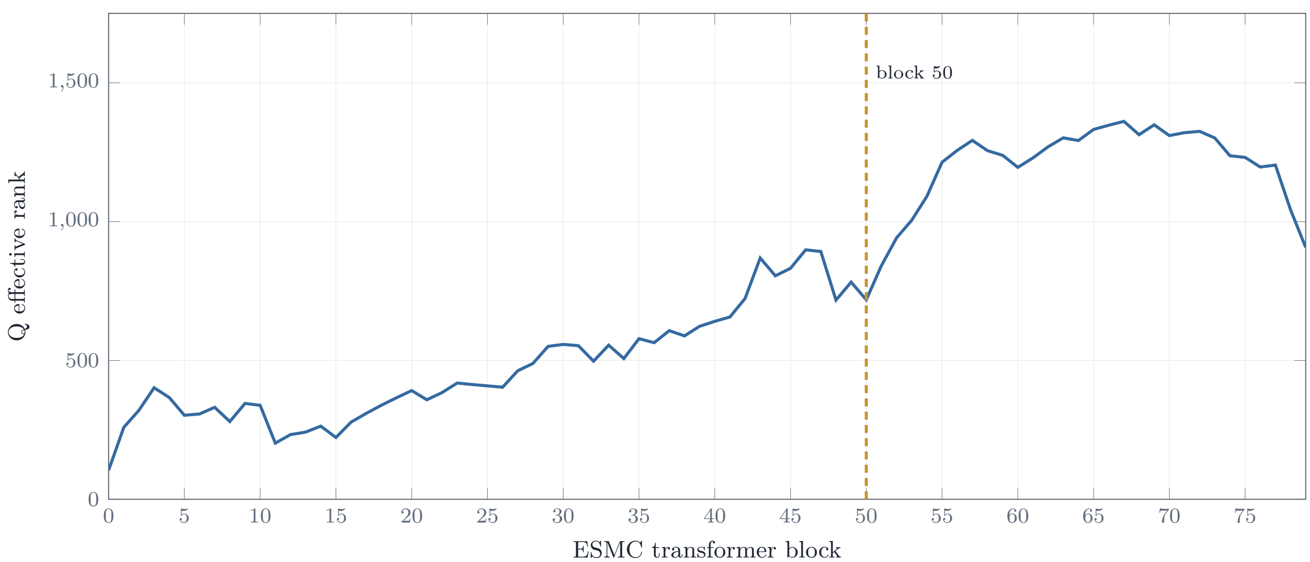

Figure 4: Entropy-based effective rank of the uncentered Q operator across depth. The global depth trend is large, motivating depth detrending before correlation with the ESMFold2 mixer.

At block 50, Q has effective rank 718.51, participation ratio 175.08, stable rank 17.23, leading energy fraction 0.0580, and spectral norm 25.95. The high effective rank means Q is not globally low-rank. The much lower stable rank means that a small number of leading singular directions dominate operator norm. A compact downstream readout can therefore select meaningful leading directions without the whole matrix being strongly compressible.

After removing the smooth depth trend, mixer mass is inversely associated with Q participation ratio ($r \approx -0.757$), Q effective rank ($r \approx -0.672$), K effective rank ($r \approx -0.675$), and Q stable rank ($r \approx -0.561$). Q effective-rank correlations at lags $-2, -1, 0, +1, +2$ are $-0.567, -0.652, -0.672, -0.554, -0.489$. The prespecified state-to-producer alignment is strongest.

8 Weight-vector intrinsic dimension

Rows and columns were treated as point clouds in raw Euclidean and unit-normalized cosine-equivalent geometry. For nearest-neighbor distances $T_1 < T_2 < \cdots$, TwoNN uses $\mu = T_2/T_1$ and

$$-\log[1 - F(\mu)] = d \log \mu. \tag{13}$$

The local Levina–Bickel estimator is

$$\widehat{d}_i(k) = \left[\frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{T_k}{T_j} \right]^{-1}, \tag{14}$$

evaluated at $k = 10, 20, 50$ together with neighborhood-distance concentration and local-dimension dispersion.

Raw Euclidean analyses produce excursions near blocks 50–51, including a raw value-column TwoNN local $z \approx 6.9$ at block 51. These excursions disappear after unit normalization. They are therefore driven mainly by vector norms rather than angular manifold complexity and fail the prespecified normalization-sensitivity criterion.

Negative result

The state-51 phenomenon is not robustly explained by a generic expansion or collapse of weight-vector intrinsic dimension. This conclusion applies to learned weight vectors, not protein representation dimension, because no activations were analyzed.

9 Layer trajectories and the broader 48–53 transition

For each tensor family, the 80 vectorized matrices define a parameter trajectory with

$$d_F(i, j) = \|W_i - W_j\|_F, \tag{15}$$

$$\cos(i, j) = \frac{\langle \operatorname{vec} W_i, \operatorname{vec} W_j \rangle}{\|W_i\|_F \|W_j\|_F}. \tag{16}$$

Centered kernel PCA on each 80×80 layer Gram matrix gives the effective dimensions in Table 4. Every trajectory has algebraic rank 79.

Table 4: Effective dimension of the 80-point layer trajectory.

Family	Down	Gate	K	O	Q	V	Value
Effective dimension	70.16	70.32	73.24	76.85	72.47	72.46	66.32
Algebraic rank	79	79	79	79	79	79	79

The layer paths are not confined to a small global subspace. Several families increase step size entering block 50, but turning angle and curvature are not uniquely extreme. The $47 \to 48$ transition ranks third by multi-family effective-rank derivative and second by stable-rank derivative. Transitions $49 \to 50$ and $50 \to 51$ rank approximately 13–14 by effective-rank derivative, with $50 \to 51$ eighth by stable-rank derivative. The defensible description is a blocks-48–53 regime containing a particularly readable state emitted by block 50.

10 Normalization, scale, and weight reconstruction

All attention and FFN LayerNorm parameters, Q/K normalization vectors, and matrix-scale statistics were examined. Metrics included moments, coefficient of variation, tail quantiles, adjacent cosine, amplified/suppressed channel fractions, channel-depth outliers, and cross-role similarity.

No normalization statistic at blocks 50 or 51 survives 5% FDR. The closest is the block-50 attention-input normalization suppressed-channel fraction at $q \approx 0.052$. Scale-invariant projection overlap and normalized-cosine checks remain positive, so ordinary matrix scale is not sufficient.

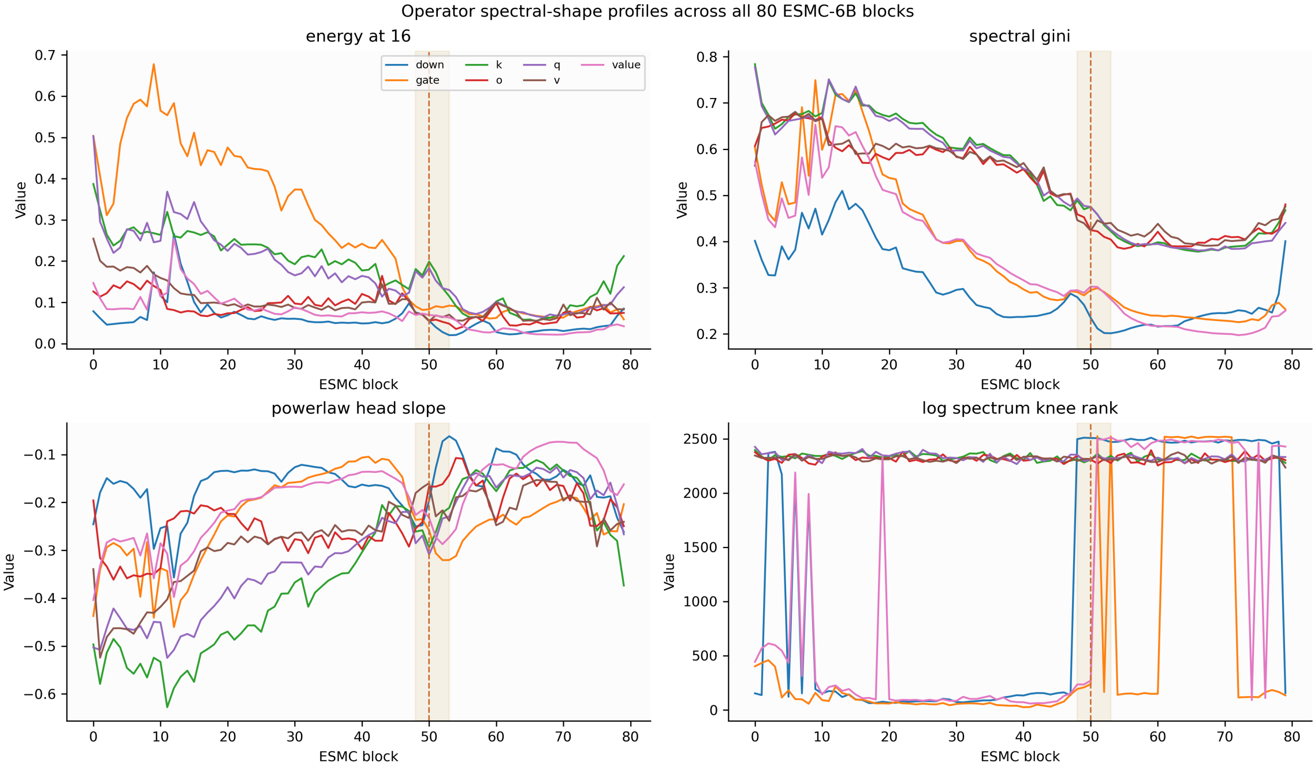

Figure 5: Operator spectral-shape profiles for Q, K, V, O, gate, value, and down matrices. The shaded region marks blocks 48–53 and the dashed line marks block 50. These profiles show a transition zone, not a unique whole-matrix rank collapse.

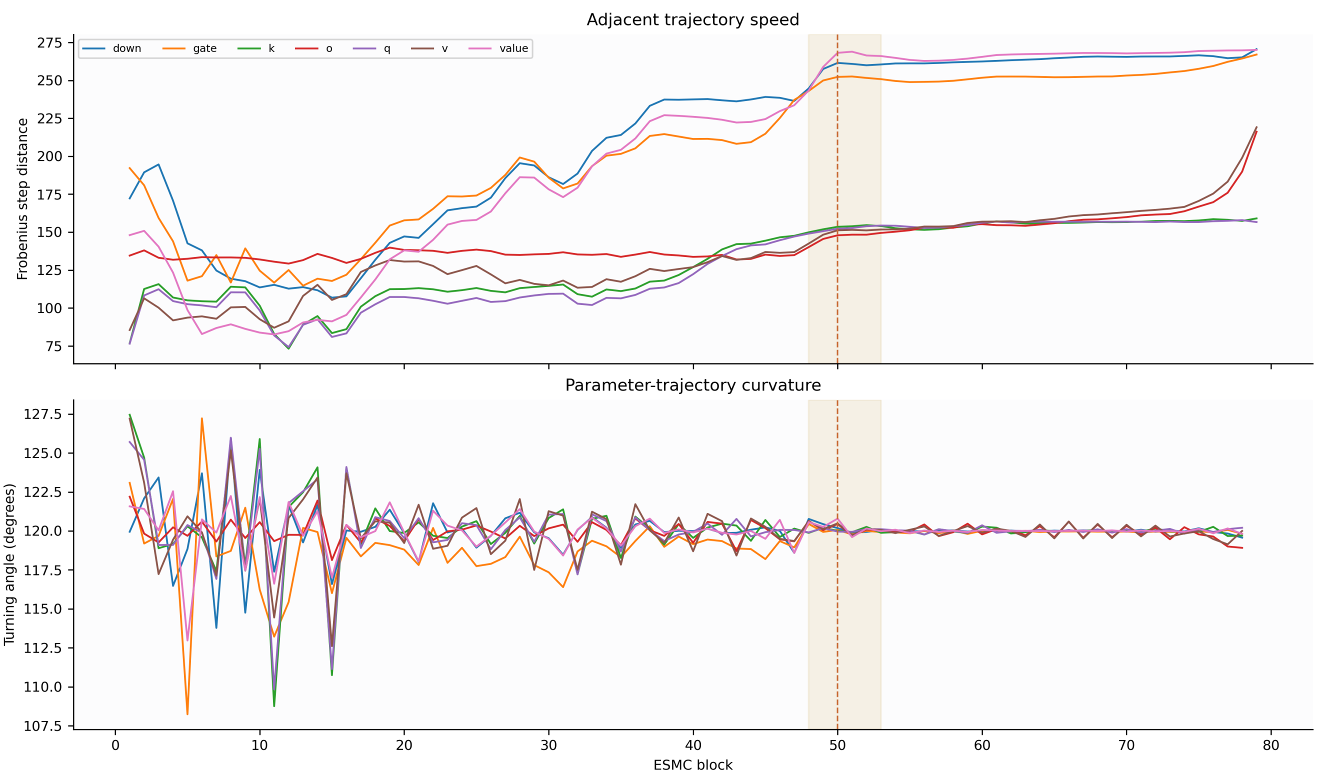

Figure 6: Adjacent Frobenius step size and parameter-trajectory turning angle. Block 50 participates in a local transition, but is not the only global trajectory change point.

Weight reconstruction included truncated SVD, symmetric per-row INT8 and INT4 quantization, unstructured magnitude sparsity at 10%, 25%, 50%, and 75%, and deterministic 2:4 sparsity. For a reconstruction $\widehat{W}$,

$$\epsilon_F = \frac{\|W - \widehat{W}\|_F}{\|W\|_F}, \tag{17}$$

$$\cos_F = \frac{\langle \operatorname{vec} W, \operatorname{vec} \widehat{W} \rangle}{\|W\|_F \|\widehat{W}\|_F}, \tag{18}$$

$$\delta_2 = \frac{|\|\widehat{W}\|_2 - \|W\|_2|}{\|W\|_2}. \tag{19}$$

No block-50 or block-51 quantization or sparsity metric survives family-wise FDR. The state is not preferred simply because its producer is uniquely compressible.

11 Layer-wise anomaly synthesis

Every metric was evaluated against a smooth depth trend, a ten-neighbor local reference, and the full depth distribution. For a target x, neighboring median m, and median absolute deviation MAD,

$$z_{\text{robust}} = 0.67448975 \frac{x - m}{\text{MAD}}. \tag{20}$$

Benjamini–Hochberg correction was applied within metric families. Discrete metrics with zero local MAD can generate infinite or extreme robust scores for a one-count change, so they were promoted only when continuous companion metrics and sensitivity checks agreed.

Table 5: Evidence summary centered on the state-51 producer.

Evidence family	Block-50 result	Multiplicity status	Interpretation
Value–projection overlap	0.15995 vs. 0.10 null	$q = 0.024$ raw; $q = 0.040$ LN-scaled	Strongest direct evidence for a folding-readable subspace
Gate–projection overlap	0.16359; local $z = 4.07$	$q = 0.132$	Supporting local peak, not a strict family-wise discovery
Attention V–O coupling	10 heads > 0.5; 6 heads > 0.7	$q = 0.008$; $q = 0.004$	Distributed multi-head event
FFN neuron coupling	gate–down mean -0.008076; broad tails	$q = 0.008$ and companion dispersion $q \leq 0.010$	Redistribution rather than simple amplification
Weight-vector dimension	Raw-only excursions	Fails unit-normalized sensitivity	Norm-driven, not robust
Normalization and scale	No 5% FDR result	closest $q \approx 0.052$	Not sufficient
Quantization and sparsity	No 5% FDR result	null	Not uniquely compressible

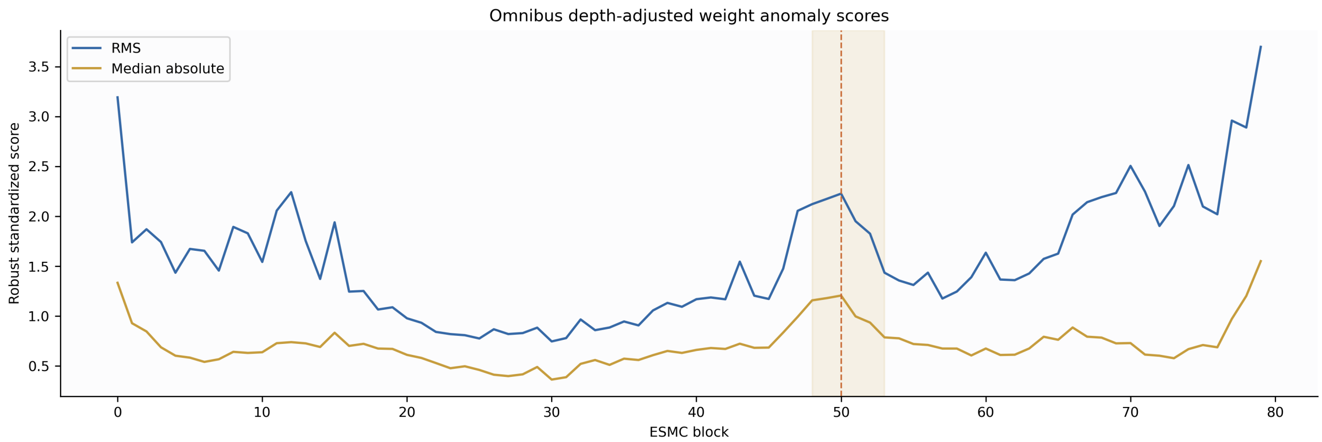

Figure 8: Omnibus depth-adjusted anomaly scores. The blocks-48–53 region is elevated, with block 50 locally prominent, but late blocks also contain strong global weight anomalies.

12 Statistical and numerical validation

Mixer associations used Pearson and Spearman correlation, cubic depth detrending, lags −2 through +2, and circular-shift nulls. With 80 blocks, an exact nonzero-shift test has 79 null alignments, so its minimum attainable p-value is $1/80 = 0.0125$. Broader 10,000-draw phase-randomized tests were used for specified metric families.

Exact Gram-spectrum calculations accumulated the smaller Gram matrix in FP64. Blocks 0, 50, 51, and 79 were checked against direct FP32 SVD. The maximum relative discrepancy among the leading 64 singular values was 4.48×10^{-4}; the maximum captured-energy discrepancy was 6.29×10^{-4}; and 90%, 95%, and 99% energy-rank thresholds differed by at most one. Negative Gram eigenvalue mass was negligible. For the block-50 down matrix it was 7.17×10^{-19}, or 5.36×10^{-20} of Frobenius-squared mass.

Independent randomized rank-64 bases were less stable: the worst leading-spectrum difference was 10.2%, the median difference was 3.0%, and the minimum replicate subspace overlap was 0.526. Inferential claims therefore rely on exact Gram/direct-SVD quantities. Randomized tail bases are retained as supporting diagnostics only.

Multiplicity and checkpoint dependence

Thousands of spectral and geometric metrics are correlated, so isolated exploratory p-values should not be treated as independent discoveries. The four ESMFold2 checkpoints share architecture and training lineage. Their agreement is a robustness check, not four independent experimental replicates.

13 Integrated interpretation

13.1 Hypothesis 1: block 50 writes a folding-readable residual subspace

Confidence: moderate; strongest weights-only hypothesis. The FFN gate/value subspaces at block 50 align with the shared folding projection, the association tracks the mixer across depth and all four checkpoints, and the value result survives raw and LayerNorm-scaled sensitivity checks. This directly explains how the folding model can prefer state 51 even when a contact-oriented probe does not.

13.2 Hypothesis 2: the write is distributed across heads and consolidated by the FFN

Confidence: moderate. Ten high-coupling V–O heads, six extreme heads, elevated O stable rank, Hungarian head reorganization, and FFN coupling dispersion coincide at block 50. Weak per-head projection significance favors an ensemble effect rather than a single folding head.

13.3 Hypothesis 3: state 51 is a transient interface state

Confidence: moderate-low. State 51 may be a convenient basis that block 50 emits and block 51 immediately transforms. This explains why the producer is unusual, the consumer is less so, and generic representational metrics can dip at the same depth. A downstream mixer can select a transient interface even when it is not the best standalone state.

13.4 Hypothesis 4: P@L and SAE metrics miss low-volume task-specific directions

Confidence: plausible but unestablished by weights. A low-dimensional direction can matter strongly to the folding projection while contributing little to average reconstruction loss, masked-LM perplexity, or contact precision. The spectra and subspace overlaps make this geometrically plausible, but only activation interventions can show that protein sequences use it.

13.5 Alternative explanations

- Checkpoint agreement may partly reflect shared initialization, data, or training lineage.
- Mixer coefficients can compensate for projected-state correlations and cancellations; a large coefficient is not unique attribution.
- The folding projection may align with a weight subspace that is rarely occupied by real activations.
- The structure-specific signal may be distributed across blocks 48–53, with state 51 serving as the most convenient linear readout point.

14 What the weights establish

Established	State 51 is heavily weighted in every supported checkpoint; block 50 produces it; and block 50 has unusual projection, head, and FFN geometry.
Not supported	The phenomenon is only matrix scale; block 51 is a robust isolated anomaly; one head causes the mixer mass; normalized intrinsic dimension explains it; or block 50 is uniquely compressible.
Unknown	Whether aligned directions activate on real proteins, causally improve folding, or represent biology rather than training-specific organization.

Validation rating

Ready to share with weights-only caveats. All 64 saved quality checks pass, direct SVD agrees with the Gram spectra, the strongest projection result survives LayerNorm and scale sensitivity, and null findings are preserved. Causal and biological claims remain intentionally withheld.

15 Recommended activation follow-up

The next phase should be narrow and causal:

1. Decompose the actual projected contributions $\alpha_s W N(h_s)$ on a fixed protein panel. Measure per-state norm, covariance, cancellation, and contribution to the final 256-vector.
2. At state 51, ablate only the rank-16 block-50 value subspace that overlaps the folding projection. Compare it with equal-rank orthogonal and matched-norm random controls.
3. Intervene on the block-50 heads with the highest V–O coupling, individually and jointly, to distinguish additivity, redundancy, and FFN gating.
4. Replace state 51 with controlled interpolations of states 50 and 52 after LayerNorm while holding the mixer and projection fixed.
5. Measure folding metrics together with P@L and the original SAE criteria. The hypothesis predicts a selective folding deficit larger than the change in generic contact or reconstruction scores.
6. Train or inspect SAEs separately on the projection-aligned component and its orthogonal complement.

The decisive experiment is an equal-rank, equal-norm, projection-aware ablation. If the aligned block-50 subspace is causal, removing it should hurt ESMFold2 more than removing an orthogonal block-50 subspace, even when both perturbations have similar effects on P@L or masked-LM perplexity.

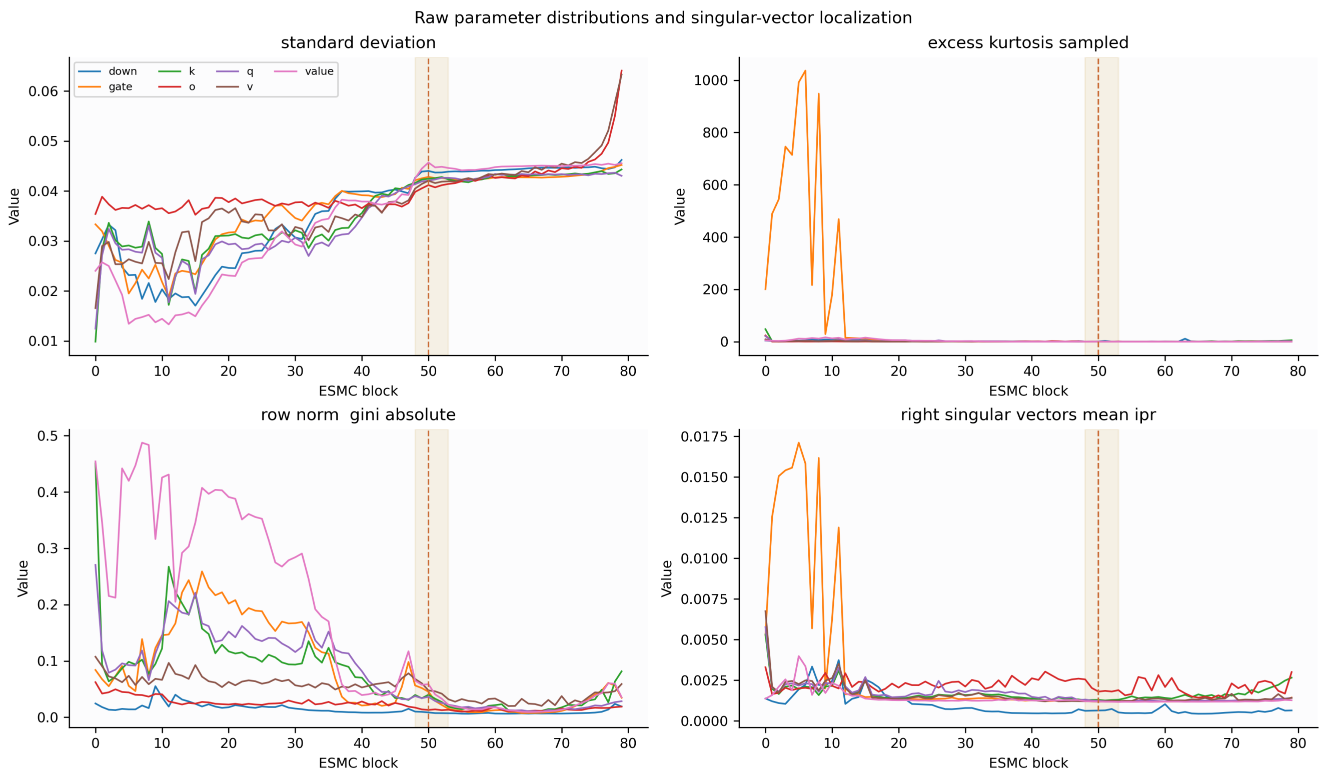

Figure 7: Raw parameter distributions and singular-vector localization. The highlighted 48–53 region does not support a simple raw-scale explanation for state 51.

A Weight inventory

The primary ESMC matrices were analyzed for every block 0–79:

Tensor family	Shape	Role
Q, K, V	2560×2560 each	Split from joint 7680×2560 QKV input projection
Attention O	2560×2560	Residual-stream attention output map
SwiGLU gate	6912×2560	First half of FFN input projection
SwiGLU value	6912×2560	Second half of FFN input projection
FFN down	2560×6912	Returns SwiGLU features to the residual stream
Attention LayerNorm	2560 weights and biases	Pre-attention channel scaling and centering
FFN LayerNorm	2560 weights and biases	Pre-FFN channel scaling and centering
Q/K norms	2560-vector parameters	Query/key normalization
Final ESMC norm	2560-vector parameters	Contextual state-80 normalization
ESMFold2 mixer	81 logits	Softmax weights over ESMC states 0–80
ESMFold2 LayerNorm	2560 weights and biases	Shared normalization preceding folding projection
ESMFold2 projection	256×2560	Shared bias-free state projection

Token embeddings and the MLM head were excluded from the primary layer analysis. Zero-length `_extra_state` metadata was ignored. The folding trunk and structure module were outside this pass.

B Analysis matrix

Lens	Measurements	Interpretation boundary
Operator spectra	Singular spectra; stable, participation, and effective ranks; energy thresholds; condition estimates	Linear operator geometry, not activation dimension
Centered PCA	Row- and column-centered spectra	Geometry of learned neuron vectors
Weight-vector dimension	TwoNN; Levina–Bickel at $k = 10, 20, 50$; concentration and dispersion	Weight-vector dimension only
Layer trajectories	Pairwise Frobenius distance, cosine, centered kernel PCA, step size, turning angle	Parameter evolution through depth
Attention heads	Norms, ranks, Q–K angles, V–O coupling, projection overlap, redundancy, Hungarian matching	Availability of head pathways, not attention usage
FFN neurons	Gate/value/down cosines, norm ratios, overlap, redundancy, concentration	Routing geometry, not neuron activation
Projection alignment	Principal angles and normalized overlap at ranks 16–256	Readable subspaces, not causal contribution
Compression	Truncated SVD, INT8/INT4, magnitude sparsity, 2:4 sparsity	Parameter reconstruction only
Anomaly testing	Depth residuals, local robust scores, global percentiles, BH FDR	Exploratory families remain dependent

C Reproducibility and provenance

The full analysis was rerun on an NVIDIA GH200 480GB workstation using CUDA 12.8 and Python 3.10.12. Checkpoint tensors were streamed with safetensors; ESMC was not instantiated or executed. Provenance records `model_execution=false`. The full remote result archive contains 765 files totaling approximately 1.17 GB at:

```
/home/ubuntu/esmc_weight_geometry_deep_dive_server_20260726
```

The pinned ESMC revision is:

```
45b0fa5d7fb06faefbd5e3b89bdcef35d564e79a
```

The six verified ESMC shard SHA-256 hashes are:

```
bd90149ff223e6ac1a0cac6147a5ae0df20d3a21df4f65356a1f19cd14f4aa8a
f75e2144d8269fe2eb4b3e0823fb089b94f176d8024153e85b8fb573a42294fa
f699f01ecc9691d9c6470492765fe54b8b5d2e9f277c139e89427433ffdfe0b2
46add1b7be098bbfdc3073884851ba3057f1b33ea23a158b650a37007dabd13d
1e1cb62f060a34e18f54a31a76683ef888b8cec59e73315f5b31d25d45a1f88c
56c73e13ae96e777ce65eee99364056069ef93b646470f352f83c5f1037b1b18
```

The four extracted ESMFold2 subset hashes are recorded in `deep_dive_provenance.json`. The server audit materializes bounded report extracts, records the exact DuckDB SQL, inventories every server artifact, and verifies all expected row counts. Every CSV field and NPZ member is documented with dtype, shape or row count, extrema, missingness, and definition.

D Evidence archive map

Artifact	Purpose
`REPORT_SOURCE.md`	Complete narrative source used to construct this document
`server_audit_summary.json`	Machine-readable synthesis of prespecified and exploratory results
`output_data_quality_checks.csv`	64 completed output validation checks, all passing
`output_field_dictionary.csv`	Field-level definitions, dtypes, row counts, null counts, distinct counts, and ranges
`output_array_catalog.csv`	NPZ member names, shapes, dtypes, sizes, extrema, and non-finite counts
`server_artifact_manifest.csv`	Inventory of all 765 server-run artifacts
`sql_validation.json`	DuckDB query execution and result-count validation
`deep_dive_provenance.json`	Hardware, software, checkpoint, stage, and command history
`report_tables/*.csv`	Bounded source tables for the primary report figures
`weights_only_deep_dive.ipynb`	Executed reproducible analysis notebook

E Open questions

- Does state 51 remain fourth-ranked if the mixer is refit while removing one state at a time, or are several states exchangeable?
- Are projection-aligned directions enriched for geometry, secondary structure, chain-interface, or confidence features?
- Do monomers, multimers, disordered proteins, and long-repeat proteins use the state-51 component differently?
- Did the mixer bump emerge early in ESMFold2 training, or only after the folding trunk learned to exploit block-50 directions?
- Does an independently initialized folding head recover the same state preference and subspace alignment?
- Is the Figure S26 trough preserved across SAE seeds, sparsity levels, normalization choices, and activation panels?

References

[1] *Language Modeling Materializes a World Model of Protein Biology.* bioRxiv preprint 2026.06.03.729735, 2026. https://doi.org/10.64898/2026.06.03.729735.

[2] Arc Institute. *Biohub ESMC-6B checkpoint*, pinned revision 45b0fa5d7fb06faefbd5e3b89bdcef35d564e79a. https://huggingface.co/biohub/ESMC-6B.

[3] FastPLMs repository. ESMFold2 language-model shim, checkpoint manifest, and weights-only analysis programs. Local reproducibility source accompanying this report.